



RAFFLES INSTITUTION
2013 Year 6 Preliminary Examination

MATHEMATICS

8864/01

Higher 1

17 September 2013

Total Marks: 95

3 hours

Additional materials : Answer Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the test, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 7 printed pages.

[Turn over

Section A: Pure Mathematics [35 marks]

- 1** Find, algebraically, the set of values of k for which

$$kx^2 + 2x + k < 0$$

for all real values of x .

[4]

- 2** Use a non-calculator method to solve the simultaneous equations:

$$\begin{aligned}y &= (2^x + 7)(2^x - 7), \\x &= 1 + \log_2(y - 11).\end{aligned}$$

[5]

- 3** Integrate

(a) $\frac{2}{6x - 7},$

[2]

(b) $3\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2.$

[3]

Find the numerical value of the gradient of the curve $y = x \ln(2x + 1)$ at the point where $x = 2$. Give your answer correct to 3 decimal places.

[2]

- 4** Consider the graph of $y = \frac{2-x}{x-k}$, where $k > 2$ is a positive constant.

Write down, in terms of k ,

(i) the equation of the vertical asymptote,

[1]

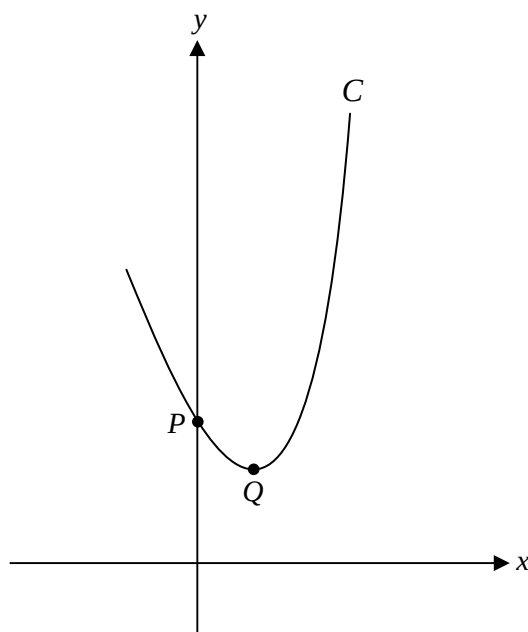
(ii) the coordinates of the intersection with the y -axis.

[1]

Sketch, on **separate diagrams**, the graphs of $y = \frac{2-x}{x-k}$ and $y = \left| \frac{2-x}{x-k} \right|$. You should indicate clearly on your graphs any points of intersection with the axes, turning points and asymptotes.

[5]

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- 5 The curve C has equation $y = ae^{2x-1} - 4x + b$, where a and b are constants. The graph of C cuts the y -axis at P and has a minimum point at Q (see diagram below).



- (i) By considering $\frac{d^2y}{dx^2}$, explain why a must be greater than zero. [2]
- (ii) The tangent at P and the normal at Q intersect at the point $\left(\frac{1}{2}, \frac{4}{e} - 1\right)$. Show that $a = 2$ and find the value of b . [6]
- (iii) Find the exact area of the region bounded by the curve C , the y -axis and the tangent at Q . [4]

[Turn over

Section B: Statistics [60 marks]

- 6** In a certain school, each student is to take up exactly one of the following three courses: Baking (B102), Knitting (K106) or Reading (R107). The numbers of male and female students in each course are given in the following table:

	B102	K106	R107
Male	40	20	220
Female	160	80	80

One of the students is chosen at random. Find, giving your answers as a fraction in its lowest term, the probability that

- (i) the student chosen is in B102, [1]
 - (ii) the student chosen is in K106, given that the student chosen is a female, [2]
 - (iii) the student chosen is a male, given that the student chosen is not in R107. [2]
- 7** The Environment ministry decides that a pilot study of a new initiative to tackle possible haze problem in schools will be undertaken before implementing the initiative nationwide. A sample of schools is to be selected to undergo this pilot study. To obtain this sample, a list of all 288 schools in the country is obtained and the schools are numbered alphabetically from 1 to 288. A number from 1 to 26 is then chosen using a random process. The corresponding school, and every 26th school thereafter, is included in the sample. For example, if the number 7 is chosen, then the schools numbered 7, 33, 59, ... , n would be included in the sample.
- (i) In the example given above, find the value of n , the number corresponding to the last school included in the sample. [1]
 - (ii) State the name given to this method of sampling. [1]
 - (iii) An advantage for this method of sampling is that it is unbiased. Explain why it is unbiased in this context. [1]
 - (iv) Give another advantage for this method of sampling in this context. [1]
 - (v) Comment on the size of the sample which would be obtained. [2]

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- 8 Events A and B are such that $P(A) = \frac{3}{7}$, $P(A \cap B) = \frac{1}{7}$ and $P(A' \cap B') = \frac{8}{21}$.

Find, in either order,

- (i) $P(B)$,
- (ii) $P(A|B')$,

leaving your answers as a fraction in its lowest term. [5]

State, giving a reason, whether events A and B are independent. [2]

- 9 The ages (x months) and the heights (y centimetres) of 10 girls are given in the table below:

x	24	100	116	120	126	144	164	178	182	192
y	88	126	141	146	145	148	155	154	156	154

- (i) Draw a scatter plot of the data. [2]
- (ii) Calculate the linear correlation coefficient between x and y . [2]
- (iii) Use an appropriate equation to estimate the height of a girl who is 26 years old. Comment on your result. [3]

[Turn over]

10 An experiment consists of throwing an ordinary fair die twice.

(i) Find the probability that the number '6' occurs at least once. [2]

(ii) By drawing a table of outcome, or otherwise, show that the probability that the sum of the scores on both throws exceeds 6 is $\frac{7}{12}$. [2]

Calculate the probability that, in 6 such experiments, there are more than 4 experiments in which the sum of the scores on both throws exceeds 6. [2]

Using a suitable approximation, calculate the probability that, in 60 such experiments, there are more than 40 experiments in which the sum of the scores on both throws exceeds 6. [4]

11 An Automated Teller Machine (ATM) is located at the West end of a certain school. Each day, the number of users, x , is recorded and the data for a particular school term (10 weeks) is summarised as follow:

Number of users, x	0	1	2	3	4	5	6	7	8	9
Frequency, f	9	6	8	20	15	8	2	1	0	1

Show that the mean for this data is 3 and calculate its variance, giving your answer as a fraction in its lowest term. [3]

In an attempt to increase the number of users, the ATM is re-located to a central location in the school. The number of users per day is recorded for another school term (10 weeks) and the new data is summarised as follow:

$$\sum (x - 3) = 28, \quad \sum (x - 3)^2 = 264.$$

Test, at the 4 % level of significance, whether the mean number of users per day has increased. State appropriate hypotheses for the test, and define any symbols used. [5]

Explain, in the context of the question, what is meant by "4% level of significance". [2]

It is given instead that the new data recorded excludes all Saturdays and Sundays. Taking the population variance to be 7, find, to 1 decimal place, the minimum level of significance that would lead to a rejection of your null hypothesis. [2]

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- 12** The time for Mr. Teo to travel to work may be taken to have a normal distribution X with mean 14 minutes and standard deviation $\sqrt{6}$ minutes.

Find the probability that, for two consecutive working days, Mr. Teo's travel time

(i) is less than 13 minutes on each day, [2]

(ii) is less than 26 minutes altogether. [2]

Explain briefly why the answer to **(ii)** is greater than the answer to **(i)**. [1]

The time for Mr. Lee to travel to work may be taken to have an independent normal distribution Y with mean μ minutes and standard deviation σ minutes.

(iii) Given that $P(Y > 13) = 0.00135$, find an equation involving both μ and σ . [2]

(iv) Given further that $P\left(Y < \frac{1}{2}(X_1 + X_2)\right) = 0.97725$, where X_1 and X_2 are two independent observations of X , calculate the values of μ and σ , giving your answers to the nearest integer. [5]

(v) Give, in the context of the question, an interpretation of $P\left(Y < \frac{1}{2}(X_1 + X_2)\right)$. [1]

— End of Paper —